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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 01/05/2007 |
| **PI Name:** | Kojima, Harry Ph.D. | | |
| **Project Title:** | Stress-Driven Instabilities on Helium-4 Crystals | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FUNDAMENTAL PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FUNDAMENTAL PHYSICS--Low temperature condensed matter physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 08854 | **Congressional District:** | 12 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-02 |
| **Start Date:** | 02/01/2001 | **End Date:** | 10/30/2006 |
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| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 2 |
| **No. of Bachelor's Candidates:** | 3 | **Monitoring Center:** | NASA JPL |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: is on NCE to 10/30/2006, per N. Hall-GRC (9/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG 3 2868 | | |
| **Performance Goal No.:** | 1.24 | | |
| **Performance Goal Text:** | To explore the frontiers of physics and uncover the mysteries of the universe obscured by Earth's gravity | | |

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| **Task Description:** | Helium-4 is pressurized at low temperatures near 1K to grow a crystal in a rectangularly shaped chamber. After a solid seed is formed, it is slowly grown to fill the lower part of the chamber. When the crystal is grown to the height of about 5 mm, the temperature and the pressure are stabilized. The crystal is then put under an externally applied non-hydrostatic stress. The external stress is applied by the use of piezoeletric piston. The interface between the liquid melt and crystal is observed as a function of the applied stress, time and temperature. Optical interferometry techniques are employed to observe the surface profile. If the applied stress is sufficiently small, the solid height is expected simply to decrease in proportion to the square of the applied stress. Interesting instabilities are expected to occur at the interface when the applied stress is large enough to exceed a certain threshold level. The instabilities under a variety of conditions of the stress direction, crystal thickness, boundary conditions and temperature are being studied.<br><br>Our objective of this project is to study the surface profile of helium-4 crystals when they are placed under externally applied stress. We wish to observe the expected corrugations of surface shape and the dynamics in the surface profile developement under different applied stress conditions and to compare these observations with predictions based on the theories of stress-driven instabilities.<br><br>The formation and growth of crystal shapes are fundamental problems in general. Helium crystals at low temperatures are chosed as the test material to study these important problems. Quantum crystals of helium-4 provide an excellent testing ground for the stress-driven instability effects owing to their high purity, temperature homogeneity, and rapid crystal growth over macroscopic sizes. The superfluid liquid melt provides the mechanisms for rapid heat dissipation, freezing and melting which allow observations of the dynamic processes over short time scales. We believe our studies of stress-driven instability can contribute significantly to better understanding of the problems. Understanding of stress-driven instability is also valuable for its close relation to the phenomena of practical importance such as pattern formation in epitaxial crystal growth, surface reaction rate on pre-stressed solids and surface cracking and corrosion. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The stress-driven instability is important in understanding of the pattern formation in epitaxial growth of nano-structures. Our research can contribute to help understand the process of self-assembly of nano-structures for the next generation nano-electronic devices. The stress-driven instability is also related to the process of cracking, which is another important problem in practical applications. Our research on instability on helium surface might lead to understanding how the cracks in solids develop. On a geological scale, certain earth formations are thought to have arizen from stress-driven instabilities. Our research on stress-driven instability with well-controlled conditions may lead to better understanding of how such geologic undulation develop under certain circumstances. |
| **Task Progress:** | A cryogenic optical interferometric apparatus for observing the surface profile of helium-4 crystal was completed and experiments on stress-driven instability on the surface of solid helium crystals were carried out. To introduce our apparatus very briefly, a He-Ne laser beam is guided into the vacuum can to illuminate the 17 mm by 10 mm cross section of the helium crystal grown in a chamber with an optical access window. A portion of the beam is partially reflected from the front window of the chamber. The rest of the beam passes through the crystal chamber, passes onto the "sample" mirror and reflected back. The two beams reflected from the front window and sample mirrors are combined to produce interference patterns. The interference image pattern is transmitted to a room temperature recording system with a CCD camera. The temporal evolution of the recorded interference patterns may finally be analyzed to determine the surface height profile of the solid-liquid interface of the helium.<br>To form solid in the chamber, the liquid pressure is raised to about 24.5 bars and the temperature is stabilized at 1.20 K. The pressure is then slowly increased until a solid seed forms. The first time a solid is grown, a pressure greater than the equilibrium melting pressure (25 bars) is usually needed before a seed appears. The presence of a solid meniscus around the seed can be seen in the image. The seed can be easily manipulated to increase or decrease in size. A movable piston is located on one side of the chamber. The pattern outside of the seed is determined by the background interference. The seed is shrunk to a small size and annealed for an hour for the purpose of removing strains and dislocations and then grown back slowly over two to three hours to a vertical thickness of 3 ~ 4 mm over the entire bottom surface of the chamber. Subsequently, stress is exerted onto the solid by displacing the piston attached to a PZT to which an external voltage Vpzt is applied. The strain produced by the motion of piston is estimated to be 6x10^(-8) times Vpzt, where Vpzt is given in the units of volts. While Vpzt is less than a threshold value (1800 ~ 2300 V), the interference pattern shows locally uneven but smooth variation of height. In general, the solid height decreases in the region near the piston. The height changes by lesser amounts or even increases in the region far away from the piston depending on the particular solid sample. According to the linear equilibrium response theory between liquid and solid under strain, the solid height is expected to fall globally. Our observations indicate that the applied strain on the solid is not uniformly transmitted across the entire length of the sample. The boundary condition between the solid helium and the chamber wall (machined copper at present) requires further studies. We have tried replacing the copper surface with glass surface in an attempt to decrease the side wall effect. So far, the side wall replacement has not substantially altered the observed effect in the small stress regime. If Vpzt does not exceed a threshold value for the sample solid, the interference pattern reverts reversibly to the original pattern when Vpzt is decreased back to zero.<br><br>When Vpzt exceeds a certain threshold value for a given sample, spectacular localized patterns develop. The localized patterns indicate large variation of the solid surface height as stress is exerted on the solid. We thus observe an onset of stress-driven instability expected on theoretical grounds. When Vpzt is increased to 4000 V, the stress induce several regions to form deep valleys whose nascent stages can be seen by the interference line pattern curling up into high line density regions. The interference pattern relaxes while Vpzt is held constant. According to theory, the stress-driven instability is not expected to relax. This unexpected interesting relaxation effects will be investigated further in the future. The interference patterns in this instability regime are no longer reversible as Vpzt is varied, and different pattern develops if the same time sequence of Vpzt is repeated. The threshold and interference pattern are also different for each solid samples. The orientation of crystal axis of our samples relative to relative to PZT is not yet controlled and it is not clear what role the crystal orientation plays. A crystal orientation effect might explain the oblique angles of the deformations produced. |
| **Bibliography Type:** | Description: (Last Updated: 01/05/2007) |

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| **Articles in Peer-reviewed Journals** | Kojima H, Grinfeld P. "Observation and simulation of stress driven instabilities on solid helium surface." American Institute of Physics Conference Proceedings. 2006 Sept;850:341-2. http://dx.doi.org/10.1063/1.2354727 , Sep-2006 |